

DR. CONNER GRIFFIN

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Curriculum Vitae

ACADEMIC APPOINTMENTS

Ross Visiting Assistant Professor
The Ohio State University

August 2025 - present

EDUCATION

Doctorate of Philosophy in Mathematics

May 2025

The University of Memphis

Dissertation: *Filter Relative Notions of Size with Applications to Semigroup Ramsey Theory*

Advisor: Randall McCutcheon

Master of Science in Mathematical Sciences

December 2022

The University of Memphis

Bachelor of Science in Mathematics

May 2019

Mississippi State University

cum laude

RESEARCH INTERESTS

I am interested in combinatorics, combinatorial number theory, and Ramsey theory. The work that I have done to date has primarily relied on techniques from general topology, logic, probability theory and semigroup theory. I have been focused on carefully defining ways of measuring sets such that “large” sets necessarily contain complex and *sparse* combinatorial configurations.

PAPERS

Publications

- *A characterization of piecewise $\mathcal{F}$ -syndetic sets.* Semigroup Forum. 109(2):375-389. 2024
<https://doi.org/10.1007/s00233-024-10468-0>

Pre-prints

- Shea D. Burns, Dennis Davenport, Shakuan Frankson, Conner Griffin, John H. Johnson Jr, Malick Kebe. *Characterizing $(\mathcal{F}, \mathcal{G})$ -syndetic, $(\mathcal{F}, \mathcal{G})$ -thick, and related notions of size using derived sets along ultrafilters.* (Submitted to Fundamenta Mathematicae). arXiv: 2503.05579
- *Infinite Sum-Product Configurations in Parallel.* arXiv: 2605.24751

FUNDING AND SUPPORT

- Van Vleet Doctoral Fellow from the Graduate School at University of Memphis (\$64,000 plus tuition waiver for four years)
- American Institute of Mathematics SQuaRE: Relative notions of size and the Stone-Čech Compactification

- TA Coach for Promoting Success in Undergraduate Mathematics (PSUM). NSF Grant Number: 1821619. (\$5000 per year)
- Fortifying open education grant supporting my authorship of Ximera course notes for Math 1130: College Algebra for Business. (\$5000)

INVITED TALKS

- **Logic Seminar**, The Ohio State University, October 2025
Title: Filter quantifiers, algebra of collections, and applications to Ramsey theory
- **Ergodic Theory Seminar**, The University of Memphis, May 2024
Title: Approaching a multidimensional Szemerédi Theorem for arithmetic trees
- **Topological Semigroups and Ramsey Theory Seminar**, Howard University, April 2024
Title: Completing a characterization of relatively piecewise syndetic sets
- **Ergodic Theory Seminar**, The University of Memphis, November 2023
Title: A new characterization of piecewise $\mathcal{F}$ -syndetic sets

SERVICE, DEVELOPMENT AND OUTREACH

The Ohio State University Spring 2026
Open Education Resources Development

- Developed complete, interactive, web-deployed course notes for Math 1130: College Algebra for Business. Provides open-source material and significantly increases curriculum alignment between this course and Math 1131: Business Calculus.
- This work is supported by a grant from the Ximera team.

The University of Memphis November 2022 - December 2024
TA Coach

- TA Coach for an NSF funded research team focused on developing pedagogical training for new graduate teachers. As the TA Coach, I performed classroom observations, led meetings to discuss effective teaching practices, and served as support and resource for the graduate student teachers in the program.

Thrive Scholars Summers 2022 – 2024
Math Professor & Teaching Assistant

- Promoted from TA (2022) to professor (2023 & 2024)
- Designed and taught a 5 week calculus intensive for high-achieving first-generation students who are transitioning to elite universities.

AWARDS AND HONORS

Graduate Assistant Meritorious Teaching Award 2024
The Graduate School, University of Memphis \$3000

- University-wide competitive award recognizing teaching effectiveness.
- One of only two recipients annually.

Dr. Ralph Faudree Research Award 2024
Department of Mathematical Sciences, University of Memphis \$500

TEACHING EXPERIENCE

The Ohio State University

August 2025 - present

Instructor of Record

- College Algebra for Business: Spring 2026
- Engineering Math A (Recitation): Fall 2025

The University of Memphis

August 2021 - May 2025

Instructor of Record

- Differential Equations: Fall 2024, Spring 2025
- Calculus I: Fall 2021, Fall 2023, Spring 2024
- Math for Elementary School Teachers I: Fall 2022

Mississippi State University

August 2017 - May 2019

Peer Tutor

- Selected for a program supporting scholarship recipients who were identified as needing tutoring in their core courses.

SELECTED REFERENCES

Dr. Randall McCutcheon (Professor)

Department of Mathematical Sciences
The University of Memphis
Memphis, TN 38152
rmcctchn at memphis dot edu

Dr. John H. Johnson (Assistant Professor)

Department of Mathematics
The Ohio State University
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